

EDUCATION

École Polytechnique, CMAP <i>CIFRE PhD Thesis:</i> Path signatures and mean-field games for modeling electricity markets and battery storage problems. <i>Supervisors:</i> Eduardo Abi Jaber, Charles Bertucci, Laurent Tur.	2024 – Present
Sorbonne University / École Polytechnique Master 2 in <i>Probability and Finance</i> . GPA: 19 / 20	2023 – 2024
Vega Institute Science Promotion Foundation Scholarship program. GPA: 5.00 / 5.00 <i>Research group:</i> Numerical experiments in rough volatility modeling, implementation of the PDE approach for local-stochastic volatility models. <i>Supervisors:</i> Charles–Henri Roubinet, Mikhail Zhitlukhin.	2021 – 2023
Moscow State University First year of Master’s program in <i>Applied Mathematics and Computer Science</i> . GPA: 5.00 / 5.00 <i>Coursework:</i> Approximation and asymptotics in the superhedging problem for a binary option. <i>Supervisor:</i> Sergey Smirnov.	2021 – 2022
Moscow State University Bachelor in <i>Applied Mathematics and Computer Science</i> (2021). GPA: 4.97 / 5.00 <i>Bachelor thesis:</i> Proximity of Bachelier and Samuelson models for different metrics. <i>Supervisor:</i> Sergey Smirnov.	2017 – 2021

WORK EXPERIENCE

PhD Quantitative Analyst, Engie Global Markets	Dec 2024 – Present
Quantitative Analyst Intern, Engie Global Markets Development and calibration of a stochastic volatility HJM model. <i>Supervisors:</i> Laurent Tur, Eduardo Abi Jaber.	Apr 2024 – Oct 2024
Scientific Coordinator, Vega Institute Science Promotion Foundation Curriculum development, assistance with the organization of conferences and schools, search for contacts and communication with lecturers and research group leaders.	Dec 2022 – Aug 2023
Quantitative Research Intern, VTB Capital Contributed to the development of a C# derivative pricing library, developed spreadsheets for risk evaluation.	Aug 2022 – Jan 2023
Expert, Center of Economic Analysis, Interfax Development and implementation of credit scoring models for small and medium-sized enterprises.	Jul 2021 – Jan 2022

AWARDS

- European Best Quant Finance Thesis Award (2024) — TopQuants & EY

PUBLICATIONS

- Abi Jaber, E., Rey, C., Sotnikov, D. (2026). *Malliavin calculus for signatures with applications to finance*. arXiv:2604.22528.
- Abi Jaber, E., Attal, E., Sotnikov, D. (2025). *Efficient Simulation of Hawkes Processes using their Affine Volterra Structure*. arXiv:2511.13554.
- Abi Jaber, E., Sotnikov, D. (2025). *Exponentially Fading Memory Signature*. arXiv:2507.03700.
- Abi Jaber, E., Gassiat, P., Sotnikov, D. (2025). *Martingale property and moment explosions in signature volatility models*. Accepted to Finance and Stochastics. arXiv:2503.17103.
- Abi Jaber, E., Bruneau, S., De Carvalho, N., Sotnikov, D., Tur, L. (2025). *Heath–Jarrow–Morton meet lifted Heston in energy markets for joint historical and implied calibration*. Accepted to Quantitative Finance. arXiv:2501.05975.

- Smirnov, S., Sotnikov, D., Zanoichkin, A. (2024). *Approximation and asymptotics in the superhedging problem for binary options*. Annals of Finance. 1-38. 10.1007/s10436-024-00454-5.
- Smirnov, S., Sotnikov, D. (2021). *Proximity of Bachelier and Samuelson Models for Different Metrics*. Review of Business and Economics Studies, 9(3), 52-76. doi:10.26794/2308-944X-2021-9-3-52-76

CONFERENCE TALKS

- April 27, 2026. *Chasing Stationarity: Exponentially Fading Memory Signature*. Stochastic Control and Games for Risk & Regulation, Hammamet, Tunisia.
- January 14, 2026. *Chasing Stationarity: Exponentially Fading Memory Signature*. 18th Colloquium Bachelier, Métabief, France.
- September 2, 2025. *Martingality in the Signature Volatility Model*. Stochastics & Computational Finance, Lisbon, Portugal.
- June 27, 2025. *Heath–Jarrow–Morton meet lifted Heston in energy markets for joint historical and implied calibration*. 12th General AMaMeF Conference, Verona, Italy.
- January 15, 2025. *Martingality in the Signature Volatility Model*. 17th Colloquium Bachelier, Métabief, France.
- December 2, 2024. *HJM Stochastic Volatility Model for Energy Markets*. Quant Careers Symposium, Amsterdam, Netherlands.

SEMINAR TALKS

- March 17, 2026. *Chasing Stationarity: Exponentially Fading Memory Signature*. Working Group “Stochastic Methods and Finance”. CERMICS, École des Ponts, Paris, France. (Invited)
- March 3, 2026. *Chasing Stationarity: Exponentially Fading Memory Signature*. DataScience@BI seminar, BI Norwegian Business School, Oslo, Norway. (Invited)
- February 24, 2026. *Chasing Stationarity: Exponentially Fading Memory Signature*. Doctoral Seminar at LPSM, Sorbonne Université, Paris, France. (Invited)
- December 10, 2025. *Chasing Stationarity: Exponentially Fading Memory Signature*. Seminar of the CMAP MathsFi group, École Polytechnique, Paris, France.
- October 4, 2025. *Chasing Stationarity: Exponentially Fading Memory Signature*. Global Seminar, Vega Institute, Moscow, Russia. (Invited, Online)
- June 5, 2025. *Probabilistic View on the Signature Method*. CEREMADE Young Researcher Days, Caen, France. (Invited)
- May 2, 2025. *Martingality in the Signature Volatility Model*. Bachelier Seminar (Doctoral Session), Paris, France.
- February 6, 2025. *Heath–Jarrow–Morton meet lifted Heston in energy markets for joint historical and implied calibration*. Young Researchers’ Seminar, Université Paris Dauphine-PSL, Paris. (Invited)
- December 11, 2024. *Martingality in the Signature Volatility Model & Stationary Signatures*. Seminar of the CMAP MathsFi group, École Polytechnique, Paris, France.

PROFESSIONAL SERVICE

Bachelier Seminar

2024 – Present

Technical management of the official website. Coordinating communications for the seminar’s global mailing list.

TEACHING & MENTORSHIP

Student Research Project (PSC), École Polytechnique

2025 – 2026

Co-supervised a group of five undergraduate students on a six-month research project regarding the application of exponentially fading memory signatures to time-series modeling.

Research Internship Supervision, Engie Global Markets

2025

Student: Mohammed Taha Belgnaoui (Master 2)

Topic: Markovian Approximation of the Volterra iVi Scheme.

Teaching Assistant, Vega Institute

2022 – 2023

Led problem-solving sessions for graduate-level courses:

Advanced Monte Carlo Methods and Applied Quantitative Finance II.

ADDITIONAL INFORMATION

IT Skills

Python (numpy, scipy, pandas, scikit-learn, sympy), C++, R, Matlab.

Languages

English (fluent), French (fluent), Russian (native)